

Lecture 1: Introduction

SOC612: Data Analytics for Social Research

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September 8, 2026

Hi everyone!

You can call me Raphi.

I have studied and lived in Zurich before coming to Hong Kong. I'm interested in public health and health policy, medical innovations, healthcare, and methodology in social sciences. I did my PhD at CUHK.

You can always write me an email (raphaelduerr@ln.edu.hk) or message me on WhatsApp (+852-6501-5091), if you have problems with an assignment, if you have a question, or when you're late for class.

Today's schedule

- 1 Welcoming!
- 2 Course Structure of SOC612
- 3 **Discussion:** Why do we need data analytics in the social sciences?
Differences to related disciplines? Why should we care in the age of AI?
Why should we use R or Python?
- 4 **Coding:** Setting up the work environment
- 5 Outlook for the next week

Welcoming!



A few questions for you..

Who are you, and how would you like to be called?

What's your academic background?

Is there any topic in data analytics or sociology that interests you?

Do you have a favourite movie or book?

SOC612: Course structure

- All materials (slides, scripts, datasets) will be made available online.
- You're generally free to use the statistical program of your choice for the assignments and during lectures. However, you're strongly advised to use either **R** or **Python**.

SOC612: Course structure

Lectures

Lectures will usually have three parts:

- 1 **Relevance of the topic for the Social Sciences:** We discuss the relevance of the lecture's topics to sociology or social sciences.
- 2 **Guided coding:** We go through a full case of the lecture's topic, and establish a standardised workflow.
- 3 **Tutorial, exercise session:** Based on the workflow, students can solve in-class exercises independently in the last part of the lecture.

The lectures last until **21.30**. For those of you living in **Shenzhen**, if you need to catch the bus, you can leave on your own.

SOC612: Course structure

Lectures

- The course is not a strict coding class, instead it focuses on the ways sociology can make use of data analytics, and how a workflow looks like.
- During the lectures, we can focus on some case studies. You will also be given time, to work independently on them.
- At the beginning, we'll do a short recap of the previous week.
- Please also read the materials listed on the syllabus.

SOC612: Course structure

Requirements

- 1 **Class and tutorial participation:** Participation in in-class activities, exercises, and tutorial sessions (10%).
- 2 **Reproducible Coding Exercise and Analytical Memo:** Students have to submit a commented, reproducible script and a sociological interpretation of their results. Datasets will be provided (40%).
- 3 **Group Computational Project:** In groups of 2 to 3, students have to conduct a small research project, based on their own sociological question. More information will be given in Week 2 (30%).
- 4 **Final Presentation and Discussion Session:** In the last session, each group has to present their group computational project, and lead a short discussion, (20%).

SOC612: Course structure

Syllabus for September

- 1 **Sep 8:** Introduction
- 2 **Sep 15:** Data structures and data cleaning: Data classes, operators, variables and variable transformations, functions, loops
- 3 **Sep 22:** Data analysis and visualization I: Exploratory and descriptive analysis, plotting
- 4 **Sep 29:** Data analysis and visualization II: Inferential analyses, interpretation and presentation of results

SOC612: Course structure

Syllabus for October

- ① **Oct 6:** Data Ethics: Privacy, algorithms and bias, AI, data scraping
- ② **Oct 13:** Online data: Online experiments and web scraping: online surveys, web scraping, analysing web data
- ③ **Oct 20:** Data analysis and visualization III: Introduction to text data
(1): Introduction to text as data, use in social sciences, processing and cleaning data
- ④ **Oct 27:** Data analysis and visualization IV: Introduction to text data
(2): Descriptives and distributions, basic models, model metrics

SOC612: Course structure

Syllabus for November

- ① **Nov 3:** Data analysis and visualization V: Goodness of fit tests, model validation, model accuracy
- ② **Nov 10:** Data analysis and visualization VI: Simulated data: Use of simulated data, imputation, simulating distributions
- ③ **Nov 17:** Data analysis and visualization VII: Machine Learning Basics: Core ideas of ML, basic models
- ④ **Nov 24:** Free lesson/Repetition: Repetition lesson or topic based on students' choice. Possible topics: Survival analysis, interactive simulations, spatial analysis, quantitative image analysis

SOC612: Course structure

Syllabus for December

- 1 **Dec 1:** Group presentations and final discussion: Presentations, discussion



Data analytics in Social Sciences

A few words at the beginning (I)

Why should we care about data analytics in social sciences?

Group activity

Form groups of 2-3 students. You'll get a short description of a popular hypothesis or theory in the social sciences.

All of them are, or were once, popular beliefs, but are now seen as flawed in some ways, thanks to more careful methodological inquiry. Try to identify the shortcomings, and possible ways to improve them.

Data analytics in Social Sciences

A few words at the beginning (I)

Why should we care about data analytics in social sciences?

- ① The "Broken Windows" Theory (Criminology)
- ② The "Mozart Effect" (Psychology/Sociology of Education)
- ③ The "Critical Period" Hypothesis (Linguistics/Sociology of Language)
- ④ The "Cultural Deprivation" Theory (Sociology of Education)
- ⑤ The "Alpha Male" Myth (Social Anthropology)

Data analytics in Social Sciences

A few words at the beginning (II)

Why should we care about data analytics in social sciences?

Some observations from the examples:

- ① **Data quality matters:** Statistical controlling and weighting can only partially correct low-quality data.
- ② **Sample size matters:** Larger sample sizes are usually better than smaller ones.
- ③ **Data sources matter:** All data types (natural, observational, laboratory etc.) have their pro's and con's.
- ④ **Theory matters:** Statistical results are usually only one part of the story. We need guiding ideas and theories to make sense of the results and embed them in a bigger narrative.

Data analytics in Social Sciences

A few words at the beginning (III)

Where are the differences between data analytics for social sciences and related areas such as applied math or statistics, or data sciences?

- **Main goal:** Normally we want to test hypotheses and explain a social phenomenon.
- **Data:** Our data is often messy and complex, and dependent on the context.
- **Methods:** Statistical methods are often just a tool for us, but not the main research focus.
- **Audience:** We usually want to make our results understandable and accessible to a broader audience, incl. policymakers, students, other academics.
- **Ethics:** Our data is normally on human subjects, so ethical questions play a bigger role.

Data analytics in Social Sciences

A few words at the beginning (IV)

Why should we learn about data analytics in the age of AI?

- **Understanding:**

To develop a deeper understanding, we can't fully rely on AI. For interpreting, assessing, evaluating, and presenting results, a good understanding is inevitable.

- **Specify AI tasks:** To specify tasks for which we use an AI, writing good and precise prompts is helpful.

- **Make informed decisions:** To make informed decisions and critically evaluate methods or changes suggested by an AI, it is helpful to start with a good understanding in data analytics.



Data analytics in Social Sciences

Working with data

Why use R or Python, when there are easier alternatives?

- **Interdisciplinary:** R/Python are used across many different subjects. Both are used outside academia.
- **Syntax:** R/Python are their own programming languages (based on some older predecessors).
- **Reproducible workflows:** It's easier to make your work fully reproducible and transparent.
- **Ongoing development:** They are both open-source community projects.
- **Applications:** R/Python offer a huge - and increasing - number of applications.



Data analytics in Social Sciences

Working with data

Why use R or Python, when there are easier alternatives?

- **Steep learning curve**
- **Not always logical consistency:** Both are user-driven, and updates are not always implemented consistently.
- **Debugging is part of the learning process and complex:** Interpreting error messages can be difficult, and debugging can take a while.



Data analytics in Social Sciences

Working with data: R and Python

- For R users: I'll upload a raw R script (*.r*) and an R Markdown (*.Rmd*) file.
- For Python users: I'll upload a raw Python script (*.py*) and a Jupyter Notebook file (*.ipynb*).
- For *.r* and *.Rmd* files, **R-Studio** is the most convenient IDE solution.
- For *.py* files, all Python IDEs (for example, **PyCharm**, **Spyder**) can read them. For *.ipynb*, you can read them, for example, in **Google Colab** or **JupyterLab**.

Working with data: GitHub

- You'll find the files on GitHub:
<https://github.com/raphael-soc/SOC612>.
- You'll also find a GitHub + Binder option to work online in JupyterLab (for both Notebook and Markdown files).
- Binder can take some minutes to load and sometimes produces errors.

Data analytics in Social Sciences

Working with data

Some important principles about coding:

- **Transparent:** Be transparent in all your steps. Specify which tools you use (incl. AI), and how you use them.
- **Completeness:** To make a script reproducible, make sure it is complete, or, if not, state, why not. Make it clear when something has to be done to the data outside the script.
- **Add comments:** Following and reading other people's code is not always easy and can take time, as there are always multiple solutions to a task. Guide your readers through your script by adding comments.

For next week

- **Reading:** *Edelmann, A., Wolff, T., Montagne, D., & Bail, C. A. (2020). Computational social science and sociology. Annual Review of Sociology, 46(1), 61–81.*
- **Working environment:** Please check your access to the course. Also, see if you can access the materials on GitHub.
- Please download R and RStudio if you plan to work with R, or any Python IDE. (<https://rstudio-education.github.io/hopr/starting.html>)
- **Lecture:** Please go through the slides again, and see if you have any questions.
- Next week, we'll go through some basic operations, preparations, and data cleaning.

Thanks for coming!

